

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

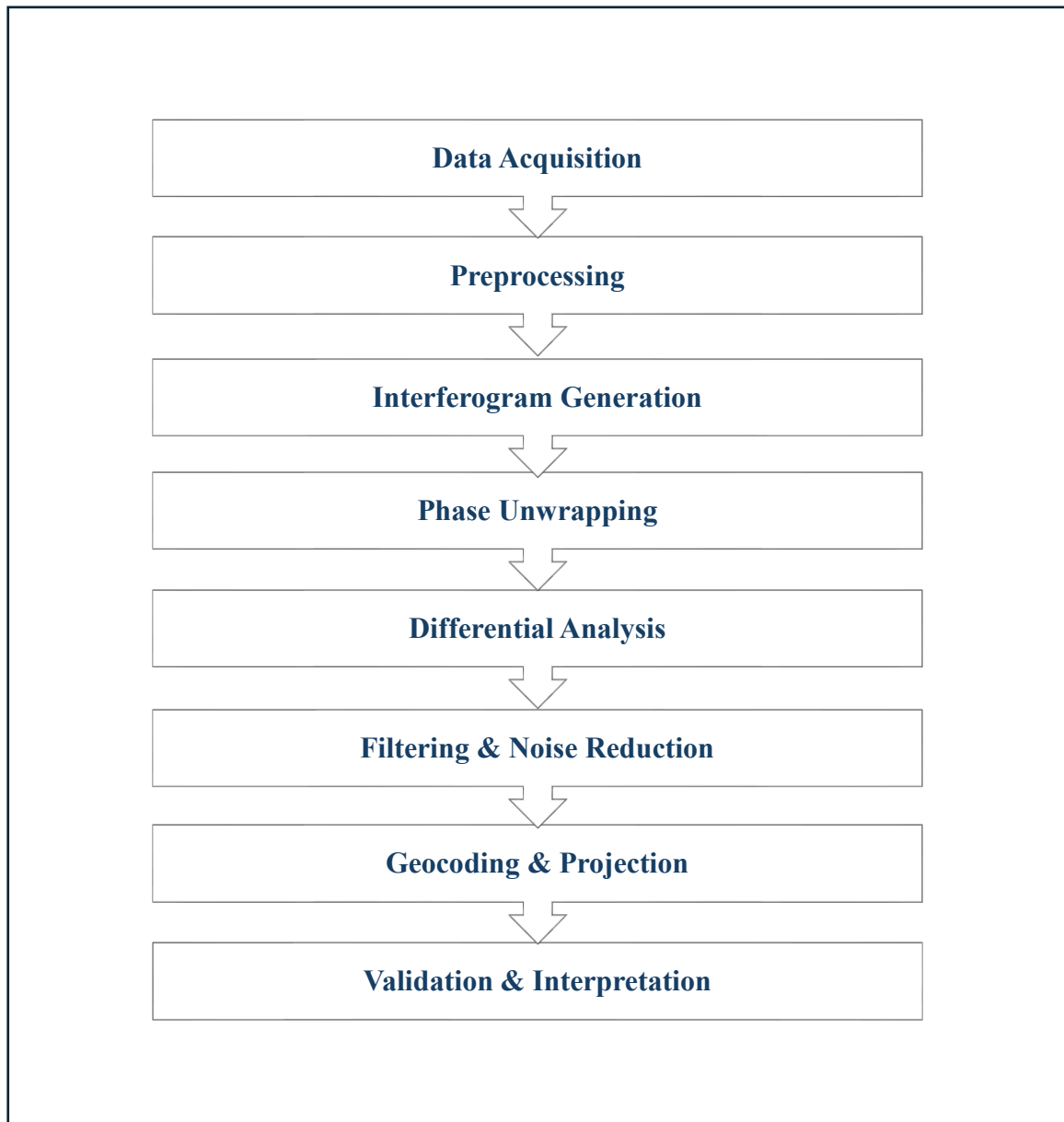
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
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S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 1015+300

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

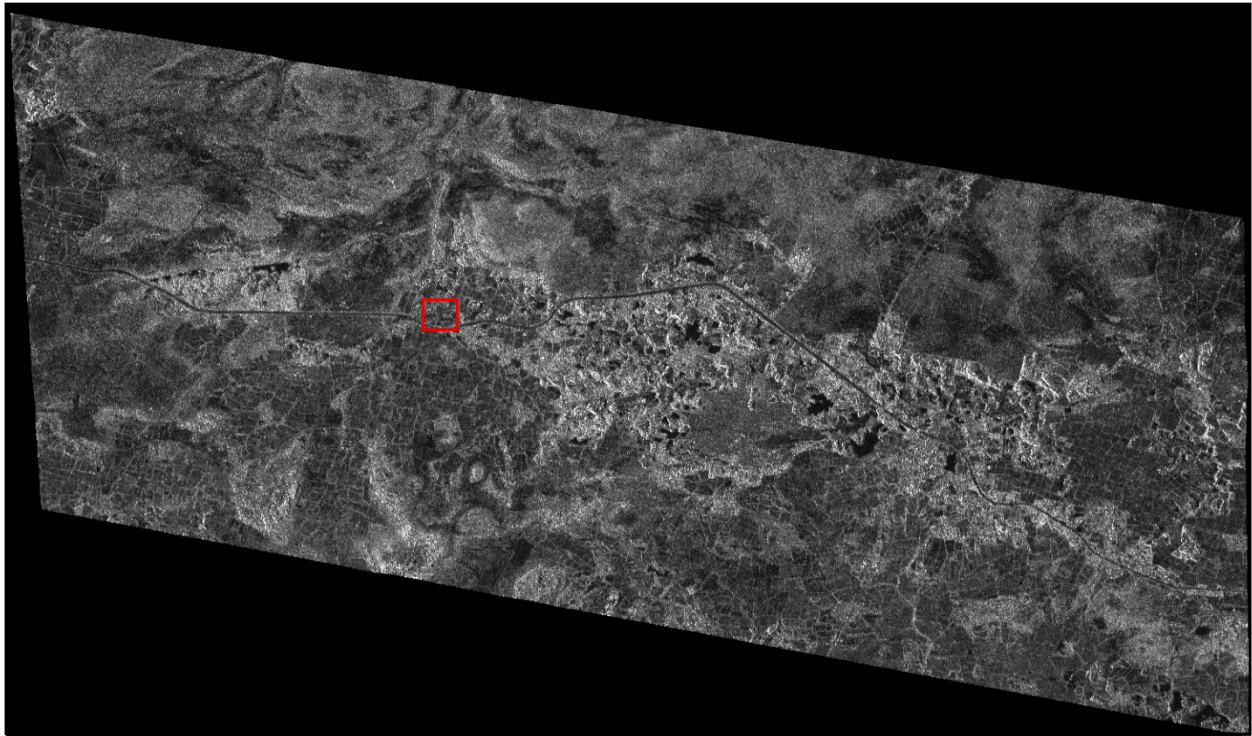
Flight Direction - DESCENDING

Polarization – VV+VH

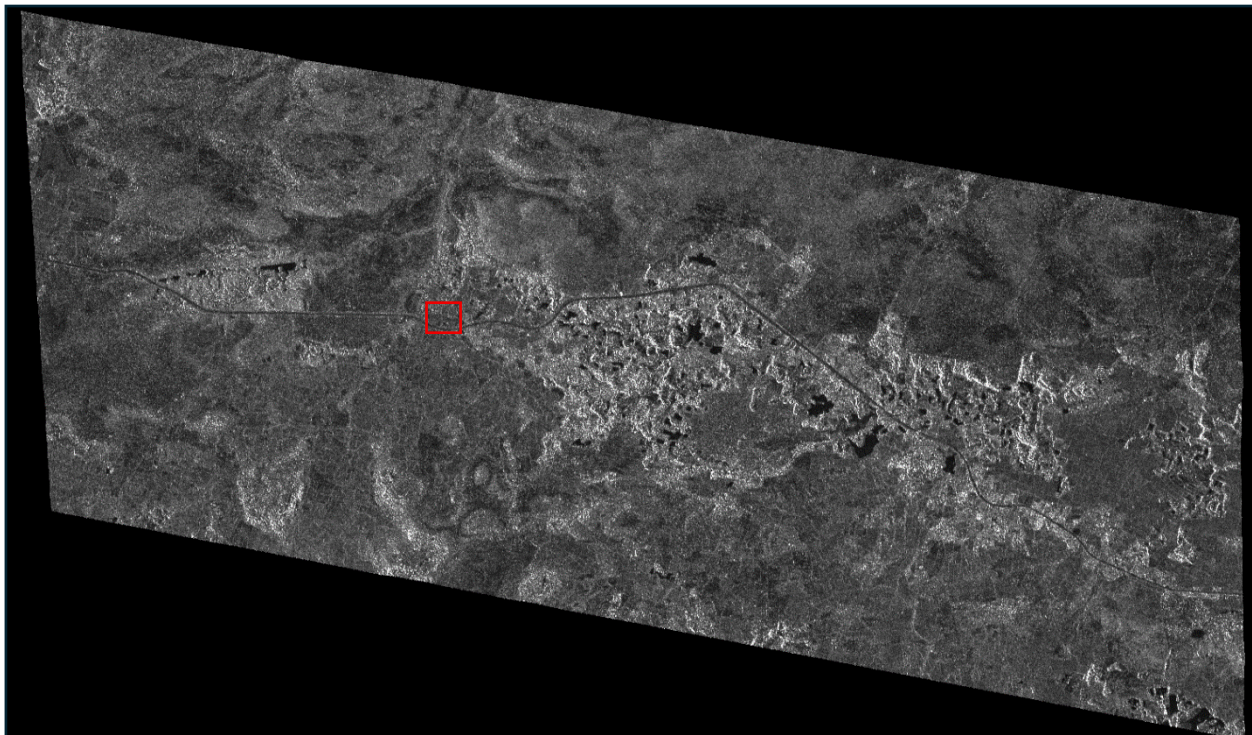
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

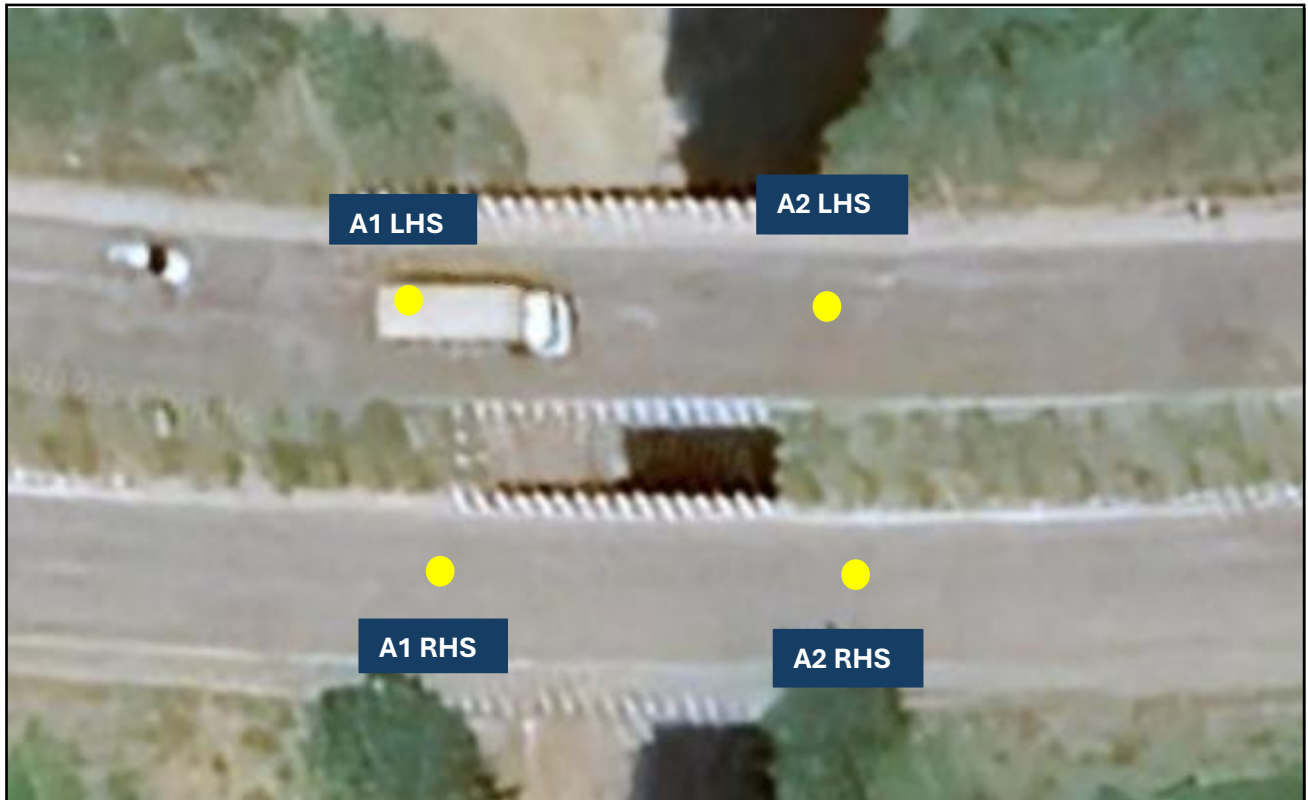


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

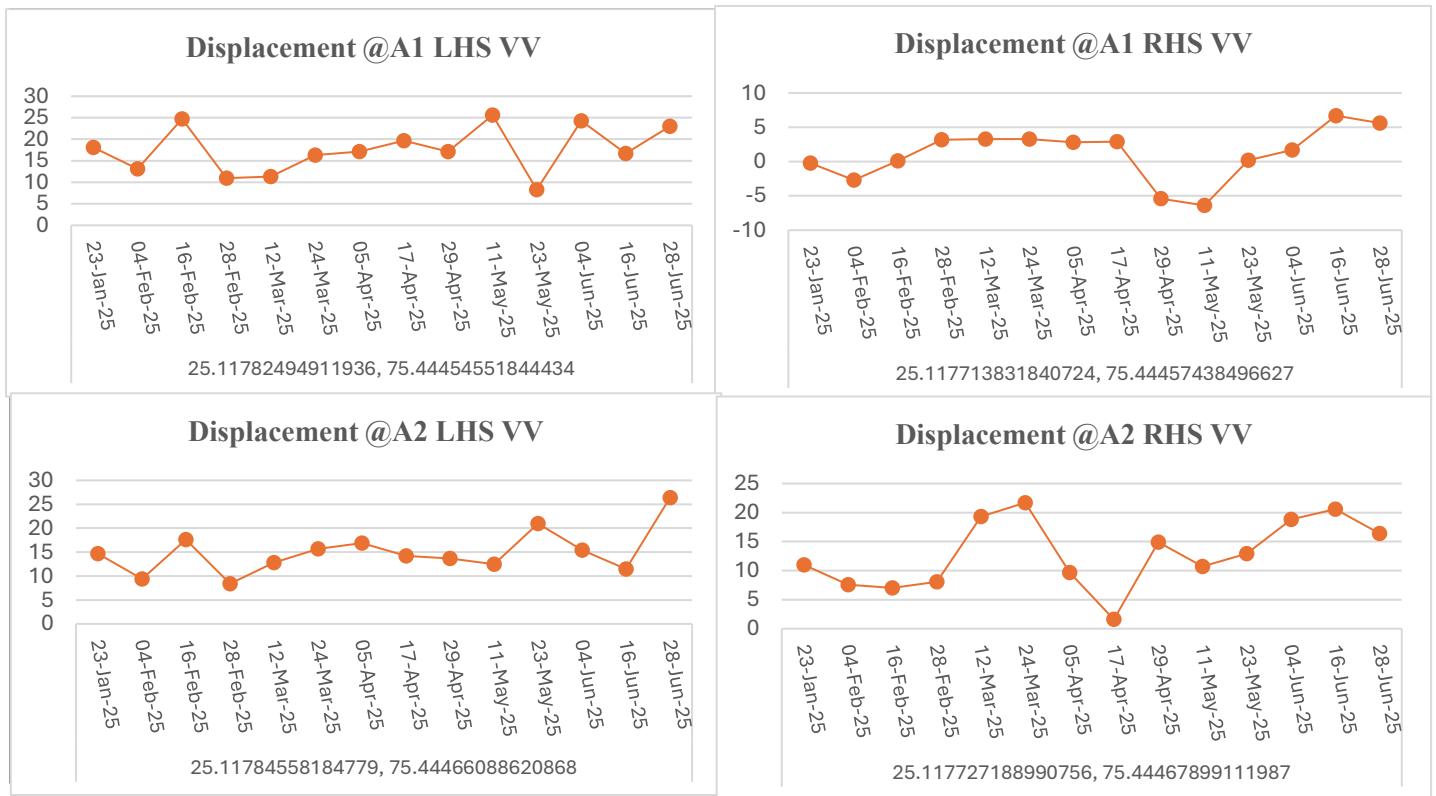
MNB – CH-1015+300

Length – 27.66M

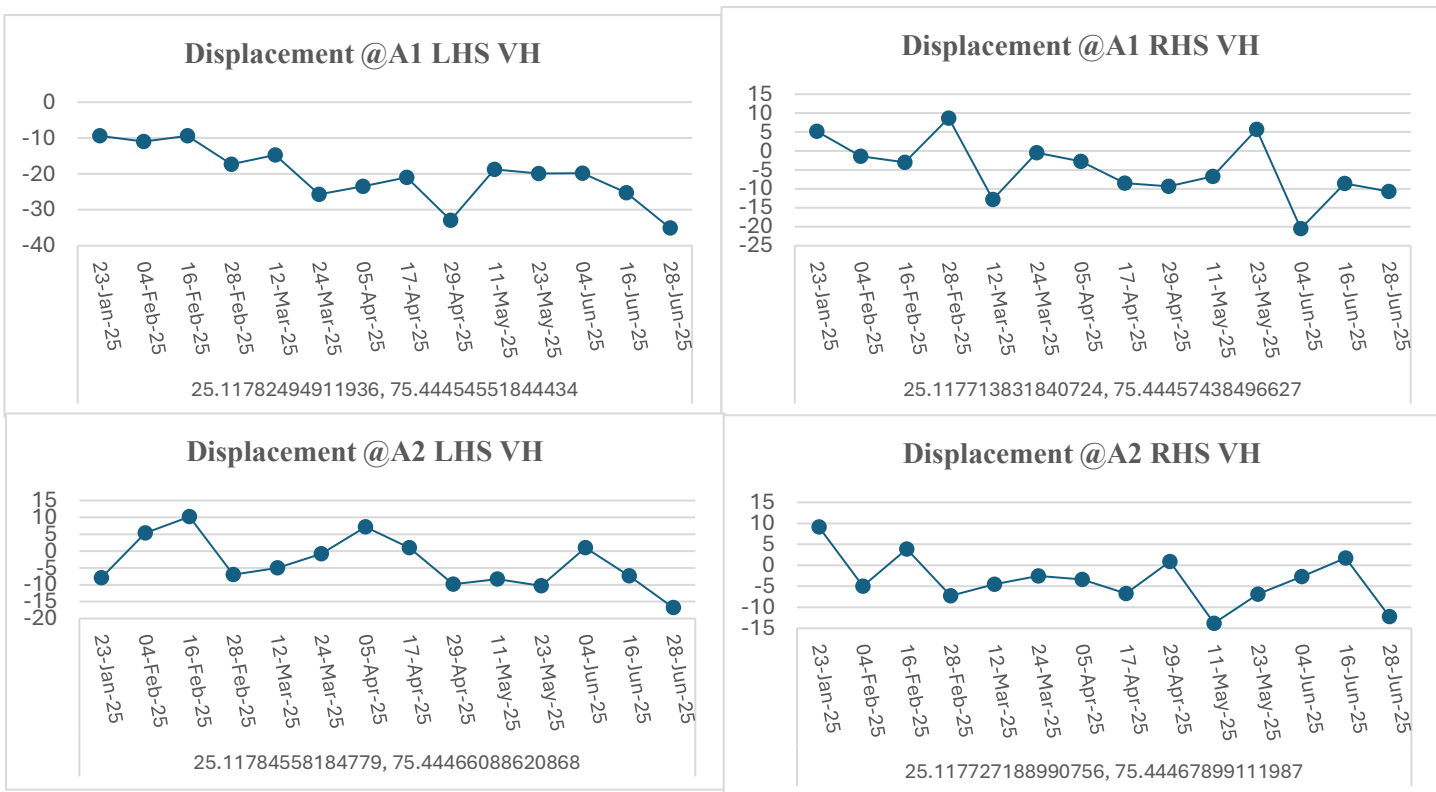


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

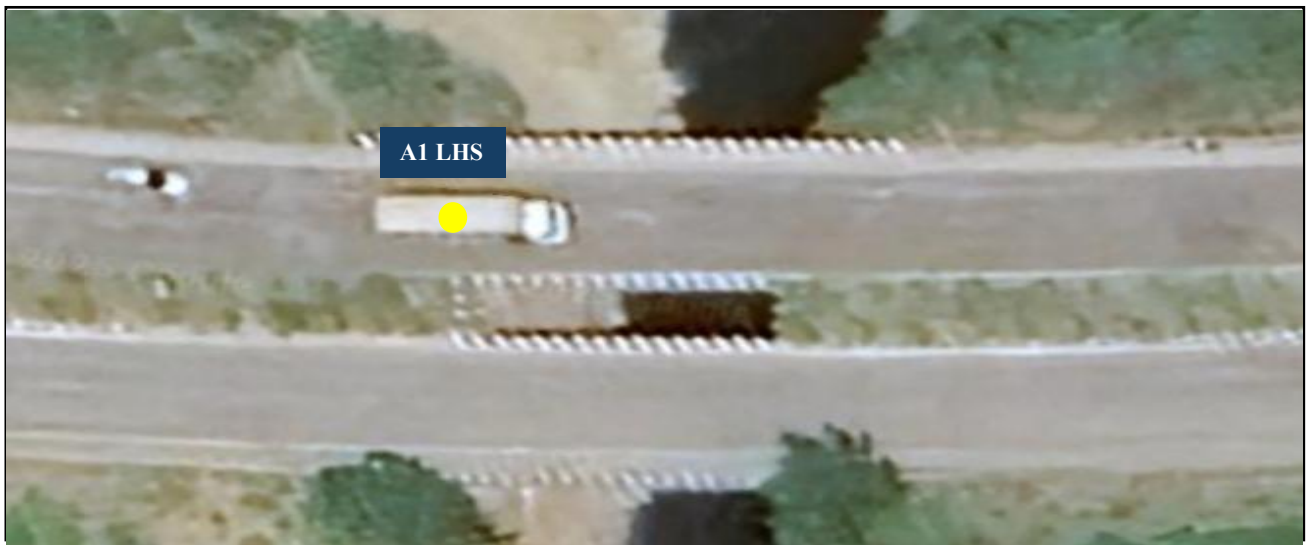
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



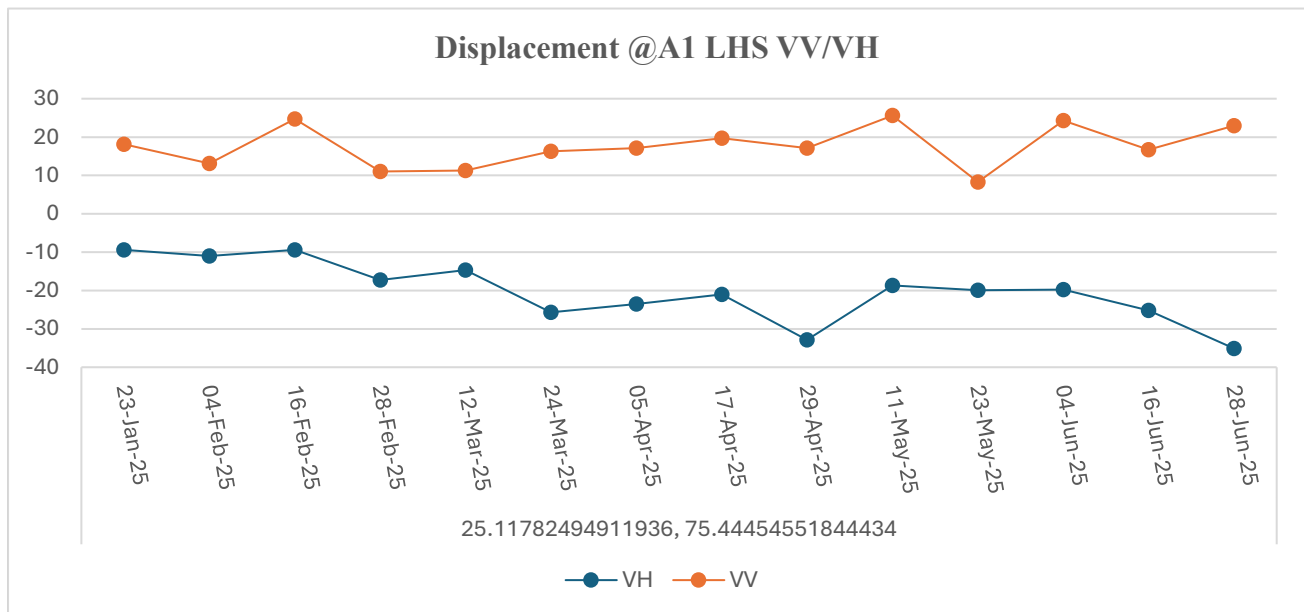
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.11782494911936, 75.44454551844434)



Key Observations:

Displacement Magnitude Patterns:

- Vertical Displacement (VV)** remains consistently positive throughout the monitoring period, generally ranging from +9 mm to +26 mm, indicating continuous upward movement. The displacement shows a gradual increasing trend, with higher magnitudes observed from April to June 2025.
- Horizontal Displacement (VH)** remains consistently negative, ranging approximately from -10 mm to -35 mm. The magnitude of displacement increases progressively with time, indicating sustained lateral movement in a single direction rather than cyclic behaviour.

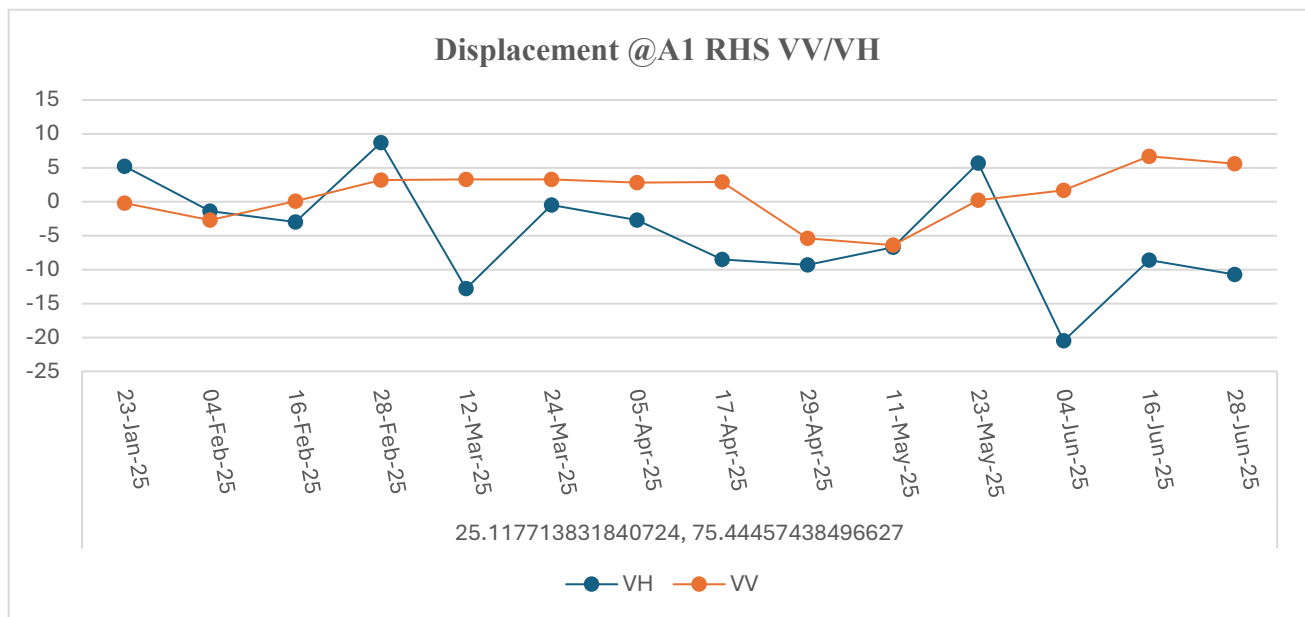
Critical Events & Anomalies:

- February-March 2025:** VH shows a marked increase in magnitude, shifting from approximately -15 mm to -25 mm, while VV remains relatively stable between +11 mm and +16 mm.
- June 2025:** VH records its maximum displacement (-35 mm) while VV remains elevated between +22 mm and +25 mm. This phase represents simultaneous high vertical uplift and extreme lateral movement, indicating the most critical deformation condition observed during the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.117713831840724, 75.44457438496627)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** exhibits high variability, ranging from approximately +9 mm to –20 mm. The displacement pattern shows frequent directional reversals, indicating cyclic lateral movement rather than a sustained unidirectional trend.
- **Vertical displacement (VV)** shows low-magnitude fluctuations around neutral to positive values, generally ranging from approximately –6 mm to +7 mm.

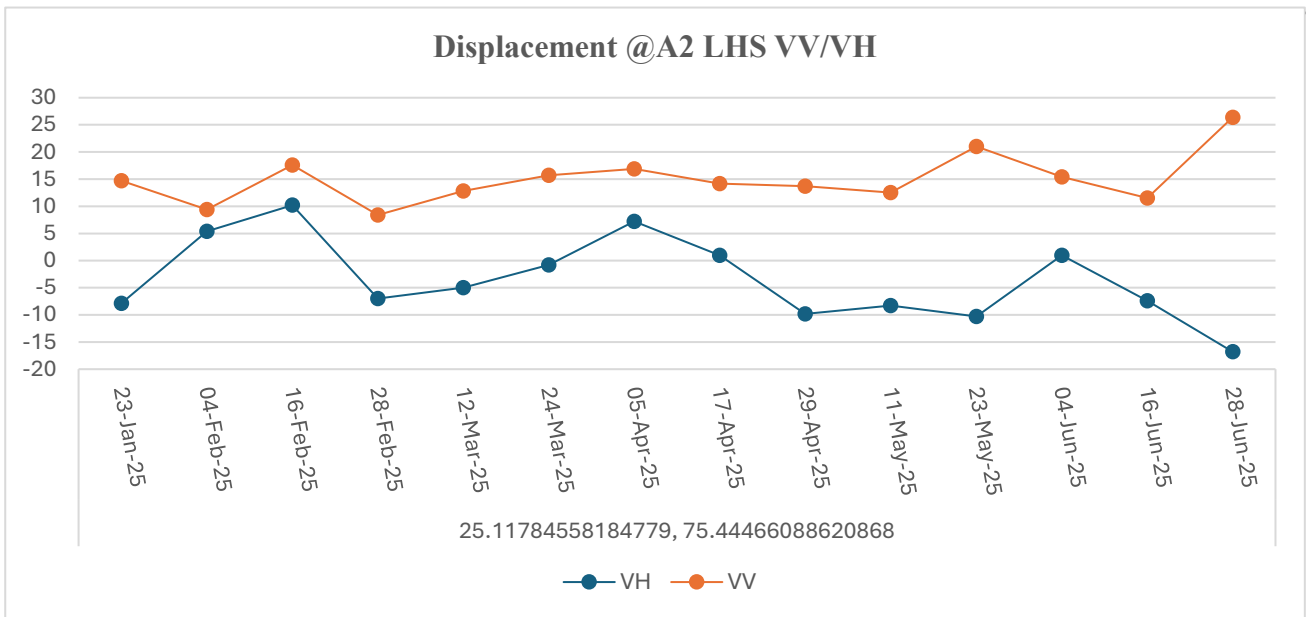
Critical Events & Anomalies:

- **March 2025:** VH drops sharply to approximately –13 mm, while VV remains stable around +4 mm. This indicates a sudden lateral displacement event with limited vertical influence.
- **June 2025:** VV increases to its highest recorded values (+6 to +7 mm) while VH partially recovers to around –9 to –11 mm, suggesting vertical uplift dominance following a major lateral movement episode.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.11784558184779, 75.44466088620868)



Key Observations:

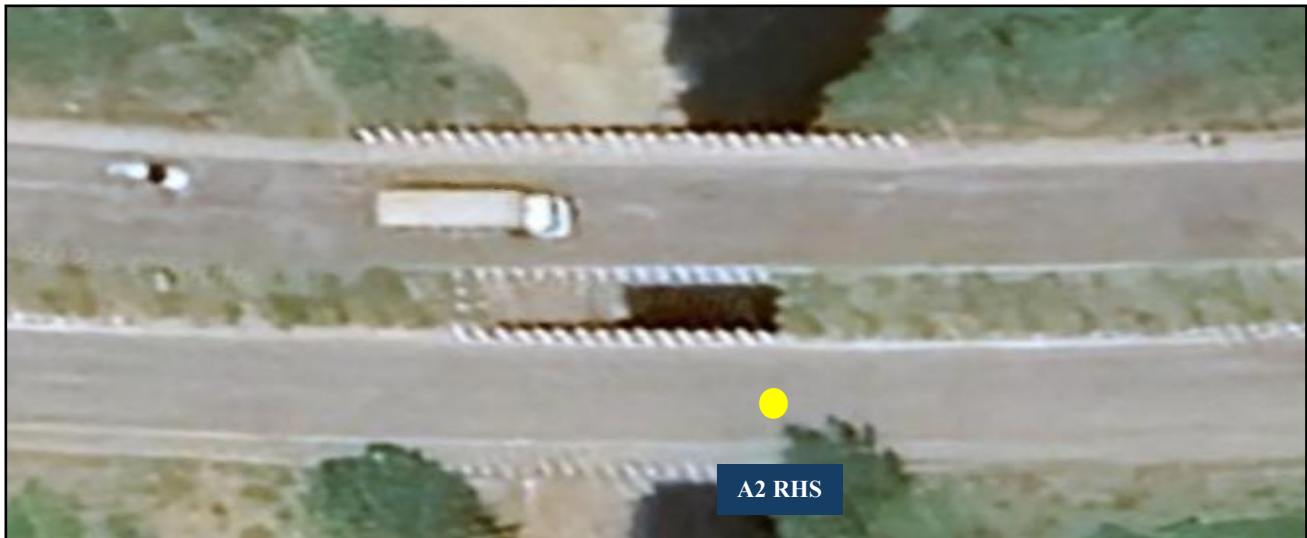
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** shows both positive and negative values, ranging approximately from +10 mm to -18 mm.
- **Vertical Displacement (VV)** remains consistently positive across the entire monitoring period, generally ranging from approximately +9 mm to +26 mm.

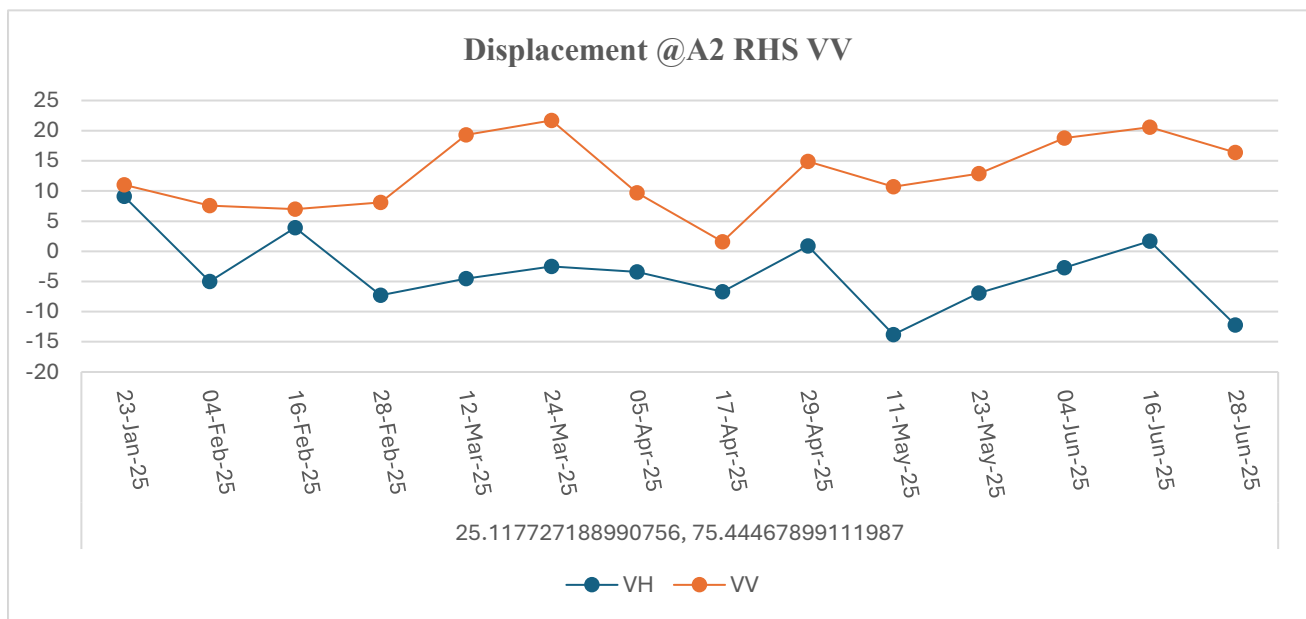
Critical Events & Anomalies:

- **February 2025:** VH reaches a local positive peak (+10 mm) while VV increases to approximately +17 mm, indicating simultaneous lateral and vertical movement.
- **April 2025:** VH records another positive excursion (+7 mm) coinciding with VV values around +16 mm, suggesting a short-term lateral movement event during continued uplift.
- **June 2025:** VH reaches its maximum negative displacement (-18 mm) while VV attains its highest recorded uplift (+26 mm).

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.117727188990756, 75.44467899111987)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** begins with negative values, indicating initial subsidence. It then follows a steady upward recovery trend, transitioning into positive displacement toward the later part of the observation period, showing gradual vertical uplift over time.
- **Horizontal (VH) displacement** displays a consistent upward progression, starting from lower positive values and increasing steadily, indicating sustained horizontal movement throughout the monitoring period.

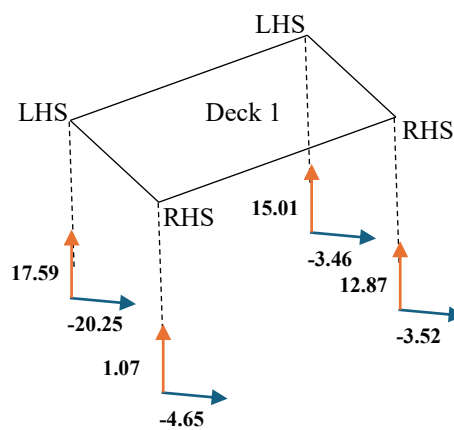
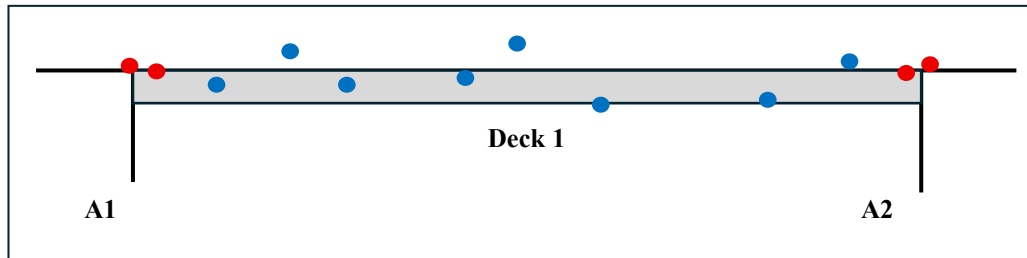
Critical Events & Anomalies:

- **February 2025:** VV remains predominantly negative while VH shows modest positive growth, suggesting initial vertical settling combined with lateral motion.
- **June 2025:** Both vertical and horizontal displacements reach their highest values, representing the most pronounced combined deformation stage, with strong uplift and horizontal movement occurring simultaneously.

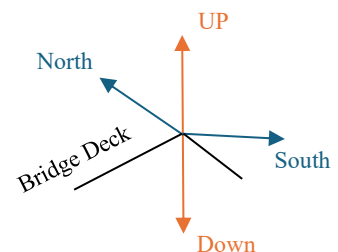
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits predominantly upward vertical displacements, with uplift values of approximately 17.59 mm at the left-hand support (LHS), 15.01 mm near the central region, and 12.87 mm toward the right-hand side (RHS).

Horizontal Displacement Behaviour:

- Horizontal displacements across the deck are moderate and spatially consistent, with movements of approximately -20.25 mm at the left-hand side, -4.65 mm near the centre, and -3.52 mm at the right-hand side, indicating a general southward shift as per the reference orientation.

CONCLUSION:

- The combined vertical and horizontal displacement patterns demonstrate that Deck 1 is behaving within expected structural performance limits, with no evidence of abnormal deformation, overstressing, or instability. Observed differential movements are consistent with normal expansion-contraction and load response mechanisms in bridge deck slabs. No corrective or remedial actions are required at this stage. Continued routine monitoring is recommended to confirm long-term stability and to identify any future deviations from these established deformation trends